

CS 162

Computer Architecture

Winter 2026

Distributed: Jan 21st, 2026

Due: 4:59pm Jan 29th, 2026

Points: 100

1. (10 pts) In practice, applying an optimization to one part of a codebase can require overheads that did not exist in the original execution or even have negative impacts on logically unrelated portions of the code. Consider a parallel code whose execution consists of SERIAL (S), PARALLEL-COMPUTE (PC), and PARALLEL-SYNC (PS) regions, where, with 1 thread, $S=20\%$, $PC=75\%$, and $PS=5\%$ of execution time. Assume that i) the amount of execution required for PS increases linearly with the number of threads, ii) the PS part is 60% parallelizable, and iii) the PC part decreases as 1:1 ratio with the number of threads. Find the number of threads that maximizes the performance of this code

Hints: Amdahl's Law

2. (10 pts) Suppose a 5-stage pipeline has the following delays for each stage

IF	ID	EX	MEM	WB
250ps	350ps	150ps	800ps	200ps

What is the clock cycle time in a pipelined and non-pipelined processor? Further assuming there are no stalls, what is the speedup achieved by pipelining datapath?

3. (20 pts) Your program consists of arithmetic instructions, load/store instructions, branch instructions, and unconditional jumps in the given ratio: 50%, 20%, 20%, and 10%. For the branch instructions, assume that 60% of all branches are Taken. Assume CPI=1 for arithmetic instructions, CPI=6 for Load/Store, base CPI=1 for branch instructions, and CPI=1 for unconditional jumps.

Your processor can support 3 control hazard strategies, and you are supposed to choose one for the given program:

- 1) Strategy 1: Unconditional stall – Stalls pipeline for 1 cycle until the branch is resolved.
- 2) Strategy 2: Branch prediction – Predicts Taken by default; Stall pipeline for 2 cycles if the branch is Not Taken.
- 3) Strategy 3: Branch prediction – Predicts Not Taken by default; Stall pipeline for 1 cycle if the branch is Taken.

Which strategy would result in the fastest runs for the given program?

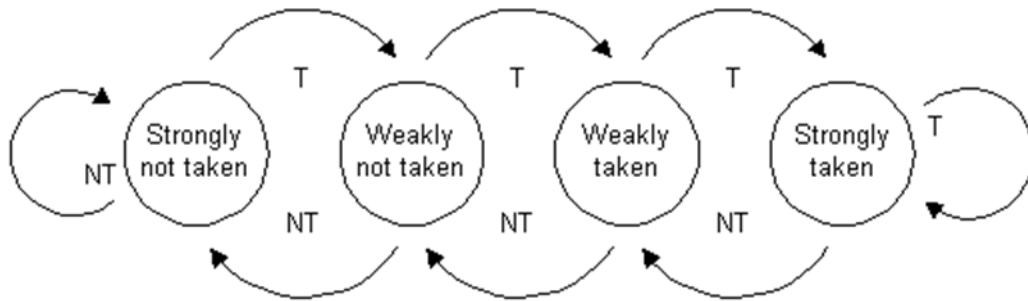
4. (30 pts) Given a program with the following repeating pattern (e.g., a loop) of branch outcomes: T, T, T, NT, NT. (T=branch taken, NT=branch not taken)

1) What is the accuracy of the 1-bit predictor for the given sequence of branch outcomes, provided we encounter this branch outcome sequence only once? Assume that the predictor starts off in the “Not Taken” state (“NT”)? **Fill the table below and calculate accuracy.**

Branch Prediction	Actual Branch behavior	Correct?
NT		
T		
T		
T		
NT		

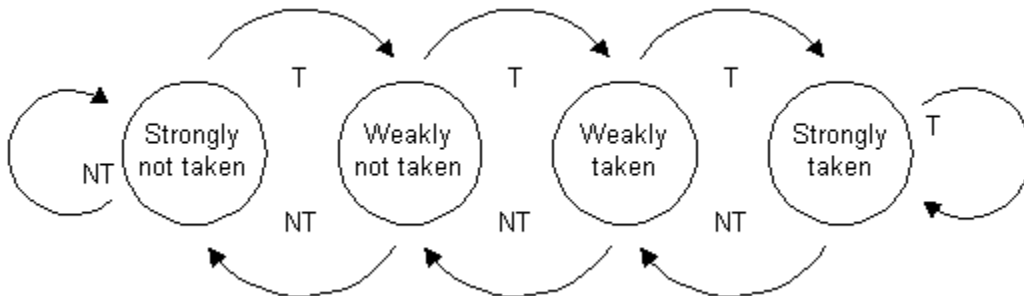
2) Using the 2-bit predictor for the given sequence of branch outcomes: T, T,NT, T, NT, what is the worst choice for the starting state (choose one from SNT, WNT, WT, and ST)?

Hint: Compute the score cards for each different starting state – SNT, WNT, WT, ST, and then compute and compare the accuracy.



3) What is the overall accuracy of the 2-bit predictor if the given pattern is repeated forever? i.e., T, T, NT, NT, NT, T, T, NT, NT, NT, to infinity and beyond?

Assume that the predictor starts off the first iteration in the “Strongly Not Taken” state (“SNT”). Subsequent iterations continue from the branch predictor state at the end of the previous iteration.



5. (30 pts) Consider the following execution sequence:

1. add \$3, \$1, \$2
2. lw \$1, 0(\$4)
3. and \$5, \$3, \$4
4. and \$6, \$1, \$2
5. or \$1, \$3, \$6
6. sw \$1, 4(\$4)
7. lw \$2, 4(\$4)

1) Find all the data hazards (assuming **no** forwarding) and fill the following table (you may leave some entries blank or add more rows to accommodate your answer).

Instruction #1	Instruction #2	Register	Hazard Type

2) Suppose the code sequence in A is executed in a 5-stage pipelined in-order execution machine. How many stalls do we need to insert into the pipeline WITHOUT a forwarding unit (your answer should be in the form of $A+B+C+\dots+Z$, where A, B, C....are the stalls inserted in the order of the instruction sequence. E.g., A is the number of stalls between instructions 1 and 2; B is the number of stalls between instructions 2 and 3, etc)?

3) How many stalls do we need to insert to the pipeline WITH a forwarding unit (NO mem to mem forwarding and the same answer format with 2)?